

Plane Kinetics of Rigid Bodies

6

CHAPTER OUTLINE

6/1 Introduction

Section A Force, Mass, and Acceleration

6/2 General Equations of Motion

6/3 Translation

6/4 Fixed-Axis Rotation

6/5 General Plane Motion

Section B Work and Energy

6/6 Work-Energy Relations

6/7 Acceleration from Work-Energy; Virtual Work

Section C Impulse and Momentum

6/8 Impulse-Momentum Equations

6/9 Chapter Review

6/1 Introduction

The *kinetics* of rigid bodies treats the relationships between the external forces acting on a body and the corresponding translational and rotational motions of the body. In Chapter 5 we developed the kinematic relationships for the plane motion of rigid bodies, and we will use these relationships extensively in this present chapter, where the effects of forces on the two-dimensional motion of rigid bodies are examined.

For our purpose in this chapter, a body which can be approximated as a thin slab with its motion confined to the plane of the slab will be considered to be in plane motion. The plane of motion will contain the mass center, and all forces which act on the body will be projected onto the plane of motion. A body which has appreciable dimensions normal to the plane of motion but is symmetrical about that plane of motion through the mass center may be treated as having plane motion. These idealizations clearly fit a very large category of rigid-body motions.

Background for the Study of Kinetics

In Chapter 3 we found that two force equations of motion were required to define the motion of a particle whose motion is confined to a plane. For the plane motion of a rigid body, an additional equation is needed to specify the state of rotation of the body. Thus, two force equations and one moment equation or their equivalent are required to determine the state of rigid-body plane motion.

The kinetic relationships which form the basis for most of the analysis of rigid-body motion were developed in Chapter 4 for a general system of particles. Frequent reference will be made to these equations as they are further developed in Chapter 6 and applied specifically to the plane motion of rigid bodies. You should refer to Chapter 4 frequently as you study Chapter 6. Also, before proceeding make sure that you have a firm grasp of the calculation of velocities and accelerations as developed in Chapter 5 for rigid-body plane motion. Unless you can determine accelerations correctly from the principles of kinematics, you frequently will be unable to apply the force and moment principles of kinetics. Consequently, you should master the necessary kinematics, including the calculation of relative accelerations, before proceeding.

Successful application of kinetics requires that you isolate the body or system to be analyzed. The isolation technique was illustrated and used in Chapter 3 for particle kinetics and will be employed consistently in the present chapter. For problems involving the instantaneous relationships among force, mass, and acceleration, the body or system should be explicitly defined by isolating it with its *free-body diagram*. When the principles of work and energy are employed, an *active-force diagram* which shows only those external forces which do work on the system may be used in lieu of the free-body diagram. The impulse-momentum diagram should be constructed when impulse-momentum methods are used. *No solution of a problem should be attempted without first defining the complete external boundary of the body or system and identifying all external forces which act on it.*

In the kinetics of rigid bodies which have angular motion, we must introduce a property of the body which accounts for the radial distribution of its mass with respect to a particular axis of rotation normal to the plane of motion. This property is known as the *mass moment of inertia* of the body, and it is essential that we be able to calculate this property in order to solve rotational problems. We assume that you are familiar with the calculation of mass moments of inertia. Appendix B treats this topic for those who need instruction or review.

Organization of the Chapter

Chapter 6 is organized in the same three sections in which we treated the kinetics of particles in Chapter 3. Section A relates the forces and moments to the instantaneous linear and angular accelerations. Section B treats the solution of problems by the method of work and energy. Section C covers the methods of impulse and momentum.

Virtually all of the basic concepts and approaches covered in these three sections were treated in Chapter 3 on particle kinetics. This repetition will help you with the topics of Chapter 6, provided you understand

the kinematics of rigid-body plane motion. In each of the three sections, we will treat three types of motion: *translation*, *fixed-axis rotation*, and *general plane motion*.

SECTION A FORCE, MASS, AND ACCELERATION

6/2 General Equations of Motion

In Arts. 4/2 and 4/4 we derived the force and moment vector equations of motion for a general system of mass. We now apply these results by starting, first, with a general rigid body in three dimensions. The force equation, Eq. 4/1,

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}} \quad [4/1]$$

tells us that the resultant $\Sigma \mathbf{F}$ of the external forces acting on the body equals the mass m of the body times the acceleration $\bar{\mathbf{a}}$ of its mass center G . The moment equation taken about the mass center, Eq. 4/9,

$$\Sigma \mathbf{M}_G = \dot{\mathbf{H}}_G \quad [4/9]$$

shows that the resultant moment about the mass center of the external forces on the body equals the time rate of change of the angular momentum of the body about the mass center.

Recall from our study of statics that a general system of forces acting on a rigid body may be replaced by a resultant force applied at a chosen point and a corresponding couple. By replacing the external forces by their equivalent force-couple system in which the resultant force acts through the mass center, we may visualize the action of the forces and the corresponding dynamic response of the body with the aid of Fig. 6/1.

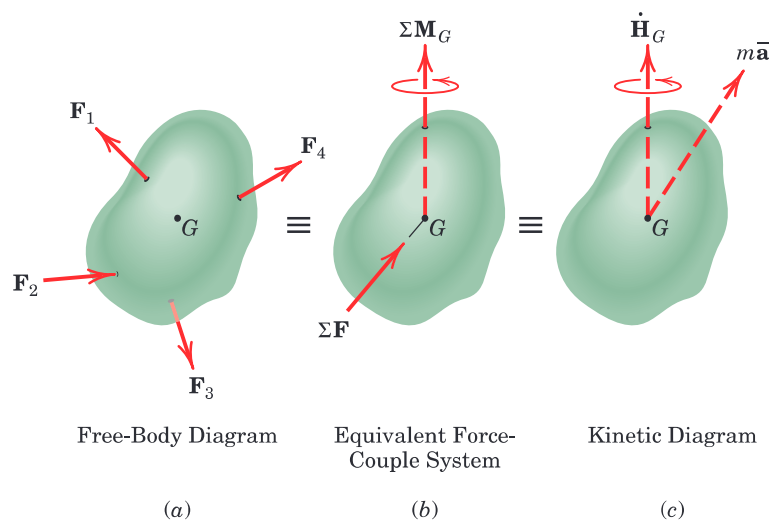


Figure 6/1

Part *a* of the figure shows the relevant free-body diagram. Part *b* of the figure shows the equivalent force-couple system with the resultant force applied through G . Part *c* of the figure is a *kinetic diagram*, which represents the resulting dynamic effects as specified by Eqs. 4/1 and 4/9. The equivalence between the free-body diagram and the kinetic diagram enables us to clearly visualize and easily remember the separate translational and rotational effects of the forces applied to a rigid body. We will express this equivalence mathematically as we apply these results to the treatment of rigid-body plane motion.

Plane-Motion Equations

We now apply the foregoing relationships to the case of plane motion. Figure 6/2 represents a rigid body moving with plane motion in the x - y plane. The mass center G has an acceleration $\bar{\mathbf{a}}$, and the body has an angular velocity $\boldsymbol{\omega} = \omega \mathbf{k}$ and an angular acceleration $\boldsymbol{\alpha} = \alpha \mathbf{k}$, both taken positive in the z -direction. Because the z -direction of both $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ remains perpendicular to the plane of motion, we may use scalar notation ω and $\alpha = \dot{\omega}$ to represent the angular velocity and angular acceleration.

The angular momentum about the mass center for the general system was expressed in Eq. 4/8a as $\mathbf{H}_G = \Sigma \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$ where $\boldsymbol{\rho}_i$ is the position vector relative to G of the representative particle of mass m_i . For our rigid body, the velocity of m_i relative to G is $\dot{\boldsymbol{\rho}}_i = \boldsymbol{\omega} \times \boldsymbol{\rho}_i$, which has a magnitude $\rho_i \omega$ and lies in the plane of motion normal to $\boldsymbol{\rho}_i$. The product $\boldsymbol{\rho}_i \times \dot{\boldsymbol{\rho}}_i$ is then a vector normal to the x - y plane in the sense of $\boldsymbol{\omega}$, and its magnitude is $\rho_i^2 \omega$. Thus, the magnitude of $\mathbf{H}_G$ becomes $H_G = \Sigma \rho_i^2 m_i \omega = \omega \Sigma \rho_i^2 m_i$. The summation, which may also be written as $\int \rho^2 dm$, is defined as the *mass moment of inertia* $\bar{I}$ of the body about the z -axis through G . (See Appendix B for a discussion of the calculation of mass moments of inertia.)

We may now write

$$H_G = \bar{I} \omega$$

where $\bar{I}$ is a constant property of the body. This property is a measure of the rotational inertia, which is the resistance to change in rotational velocity due to the radial distribution of mass around the z -axis through G . With this substitution, our moment equation, Eq. 4/9, becomes

$$\Sigma M_G = \dot{H}_G = \bar{I} \dot{\omega} = \bar{I} \alpha$$

where $\alpha = \dot{\omega}$ is the angular acceleration of the body.

We may now express the moment equation and the vector form of the generalized Newton's second law of motion, Eq. 4/1, as

$$\begin{aligned} \Sigma \mathbf{F} &= m \bar{\mathbf{a}} \\ \Sigma M_G &= \bar{I} \alpha \end{aligned} \quad (6/1)$$

Equations 6/1 are the general equations of motion for a rigid body in plane motion. In applying Eqs. 6/1, we express the vector force equation

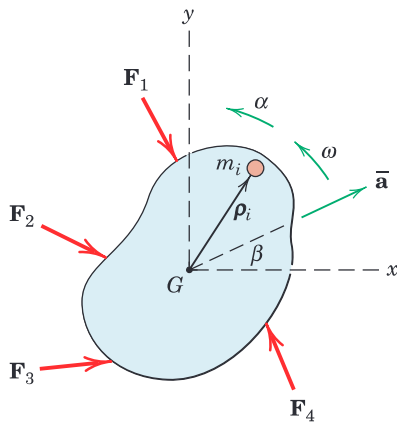


Figure 6/2

in terms of its two scalar components using x - y , n - t , or r - θ coordinates, whichever is most convenient for the problem at hand.

Alternative Derivation

It is instructive to use an alternative approach to derive the moment equation by referring directly to the forces which act on the representative particle of mass m_i , as shown in Fig. 6/3. The acceleration of m_i equals the vector sum of $\bar{a}$ and the relative terms $\rho_i\omega^2$ and $\rho_i\alpha$, where the mass center G is used as the reference point. It follows that the resultant of all forces on m_i has the components $m_i\bar{a}$, $m_i\rho_i\omega^2$, and $m_i\rho_i\alpha$ in the directions shown. The sum of the moments of these force components about G in the sense of α becomes

$$M_{G_i} = m_i\rho_i^2\alpha + (m_i\bar{a} \sin \beta)x_i - (m_i\bar{a} \cos \beta)y_i$$

Similar moment expressions exist for all particles in the body, and the sum of these moments about G for the resultant forces acting on all particles may be written as

$$\Sigma M_G = \Sigma m_i\rho_i^2\alpha + \bar{a} \sin \beta \Sigma m_i x_i - \bar{a} \cos \beta \Sigma m_i y_i$$

But the origin of coordinates is taken at the mass center, so that $\Sigma m_i x_i = m\bar{x} = 0$ and $\Sigma m_i y_i = m\bar{y} = 0$. Thus, the moment sum becomes

$$\Sigma M_G = \Sigma m_i\rho_i^2\alpha = \bar{I}\alpha$$

as before. The contribution to ΣM_G of the forces internal to the body is, of course, zero since they occur in pairs of equal and opposite forces of action and reaction between interacting particles. Thus, ΣM_G , as before, represents the sum of moments about the mass center G of only the external forces acting on the body, as disclosed by the free-body diagram.

We note that the force component $m_i\rho_i\omega^2$ has no moment about G and conclude, therefore, that the angular velocity ω has no influence on the moment equation about the mass center.

The results embodied in our basic equations of motion for a rigid body in plane motion, Eqs. 6/1, are represented diagrammatically in Fig. 6/4,

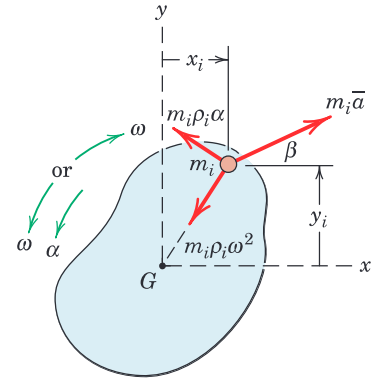


Figure 6/3

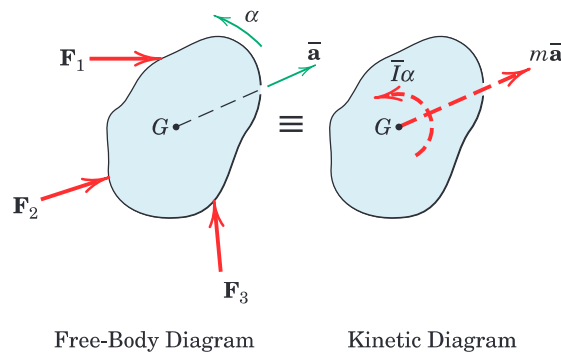


Figure 6/4

which is the two-dimensional counterpart of parts *a* and *c* of Fig. 6/1 for a general three-dimensional body. The free-body diagram discloses the forces and moments appearing on the left-hand side of our equations of motion. The kinetic diagram discloses the resulting dynamic response in terms of the translational term $m\bar{\mathbf{a}}$ and the rotational term $\bar{I}\alpha$ which appear on the right-hand side of Eqs. 6/1.

As previously mentioned, the translational term $m\bar{\mathbf{a}}$ will be expressed by its *x-y*, *n-t*, or *r-θ* components once the appropriate inertial reference system is designated. The equivalence depicted in Fig. 6/4 is basic to our understanding of the kinetics of plane motion and will be employed frequently in the solution of problems.

Representation of the resultants $m\bar{\mathbf{a}}$ and $\bar{I}\alpha$ will help ensure that the force and moment sums determined from the free-body diagram are equated to their proper resultants.

Alternative Moment Equations

In Art. 4/4 of Chapter 4 on systems of particles, we developed a general equation for moments about an arbitrary point *P*, Eq. 4/11, which is

$$\Sigma \mathbf{M}_P = \dot{\mathbf{H}}_G + \bar{\boldsymbol{\rho}} \times m\bar{\mathbf{a}} \quad [4/11]$$

where $\bar{\boldsymbol{\rho}}$ is the vector from *P* to the mass center *G* and $\bar{\mathbf{a}}$ is the mass-center acceleration. As we have shown earlier in this article, for a rigid body in plane motion $\dot{\mathbf{H}}_G$ becomes $\bar{I}\alpha$. Also, the cross product $\bar{\boldsymbol{\rho}} \times m\bar{\mathbf{a}}$ is simply the moment of magnitude $m\bar{a}d$ of $m\bar{\mathbf{a}}$ about *P*. Therefore, for the two-dimensional body illustrated in Fig. 6/5 with its free-body diagram and kinetic diagram, we may rewrite Eq. 4/11 simply as

$$\Sigma M_P = \bar{I}\alpha + m\bar{a}d \quad (6/2)$$

Clearly, all three terms are positive in the counterclockwise sense for the example shown, and the choice of *P* eliminates reference to $\mathbf{F}_1$ and $\mathbf{F}_3$.

If we had wished to eliminate reference to $\mathbf{F}_2$ and $\mathbf{F}_3$, for example, by choosing their intersection as the reference point, then *P* would lie on the opposite side of the $m\bar{\mathbf{a}}$ vector, and the clockwise moment of $m\bar{\mathbf{a}}$

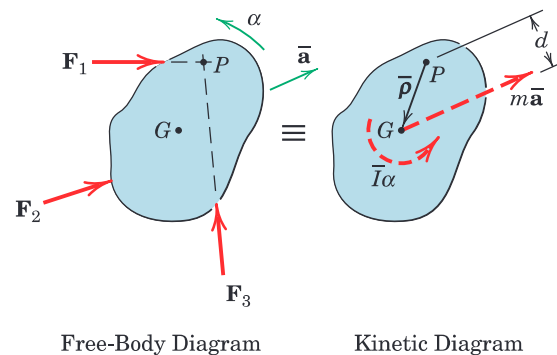


Figure 6/5

about P would be a negative term in the equation. Equation 6/2 is easily remembered as it is merely an expression of the familiar principle of moments, where the sum of the moments about P equals the combined moment about P of their sum, expressed by the resultant couple $\Sigma \mathbf{M}_G = \bar{I}\alpha$ and the resultant force $\Sigma \mathbf{F} = m\bar{\mathbf{a}}$.

In Art. 4/4 we also developed an alternative moment equation about P , Eq. 4/13, which is

$$\Sigma \mathbf{M}_P = (\dot{\mathbf{H}}_P)_{\text{rel}} + \bar{\boldsymbol{\rho}} \times m\mathbf{a}_P \quad [4/13]$$

For rigid-body plane motion, if P is chosen as a point *fixed* to the body, then in scalar form $(\dot{\mathbf{H}}_P)_{\text{rel}}$ becomes $I_P\alpha$, where I_P is the mass moment of inertia about an axis through P and α is the angular acceleration of the body. So we may write the equation as

$$\Sigma M_P = I_P\alpha + \bar{\rho} \times m\mathbf{a}_P \quad (6/3)$$

where the acceleration of P is $\mathbf{a}_P$ and the position vector from P to G is $\bar{\boldsymbol{\rho}}$.

When $\bar{\boldsymbol{\rho}} = \mathbf{0}$, point P becomes the mass center G , and Eq. 6/3 reduces to the scalar form $\Sigma M_G = \bar{I}\alpha$, previously derived. When point P becomes a point O fixed in an inertial reference system and attached to the body (or body extended), then $\mathbf{a}_P = \mathbf{0}$, and Eq. 6/3 in scalar form reduces to

$$\Sigma M_O = I_O\alpha \quad (6/4)$$

Equation 6/4 then applies to the rotation of a rigid body about a nonaccelerating point O fixed to the body and is the two-dimensional simplification of Eq. 4/7.

Unconstrained and Constrained Motion

The motion of a rigid body may be unconstrained or constrained. The rocket moving in a vertical plane, Fig. 6/6*a*, is an example of unconstrained motion as there are no physical confinements to its motion.

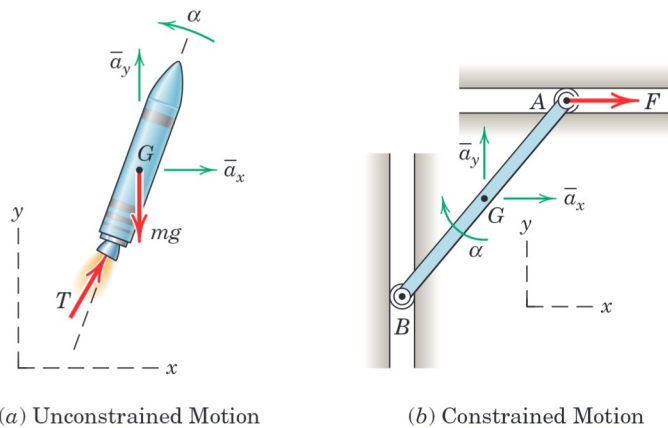


Figure 6/6

The two components $\bar{a}_x$ and $\bar{a}_y$ of the mass-center acceleration and the angular acceleration α may be determined independently of one another by direct application of Eqs. 6/1.

The bar in Fig. 6/6*b*, on the other hand, undergoes a constrained motion, where the vertical and horizontal guides for the ends of the bar impose a kinematic relationship between the acceleration components of the mass center and the angular acceleration of the bar. Thus, it is necessary to determine this kinematic relationship from the principles established in Chapter 5 and to combine it with the force and moment equations of motion before a solution can be carried out.

In general, dynamics problems which involve physical constraints to motion require a kinematic analysis relating linear to angular acceleration before the force and moment equations of motion can be solved. It is for this reason that an understanding of the principles and methods of Chapter 5 is so vital to the work of Chapter 6.

Systems of Interconnected Bodies

Upon occasion, in problems dealing with two or more connected rigid bodies whose motions are related kinematically, it is convenient to analyze the bodies as an entire system.

Figure 6/7 illustrates two rigid bodies hinged at A and subjected to the external forces shown. The forces in the connection at A are internal to the system and are not disclosed. The resultant of all external forces must equal the vector sum of the two resultants $m_1\bar{\mathbf{a}}_1$ and $m_2\bar{\mathbf{a}}_2$, and the sum of the moments about some arbitrary point P of all external forces must equal the moment of the resultants, $\bar{I}_1\alpha_1 + \bar{I}_2\alpha_2 + m_1\bar{\mathbf{a}}_1d_1 + m_2\bar{\mathbf{a}}_2d_2$. Thus, we may state

$$\begin{aligned}\Sigma \mathbf{F} &= \Sigma m \bar{\mathbf{a}} \\ \Sigma M_P &= \Sigma \bar{I} \alpha + \Sigma m \bar{\mathbf{a}} d\end{aligned}\quad (6/5)$$

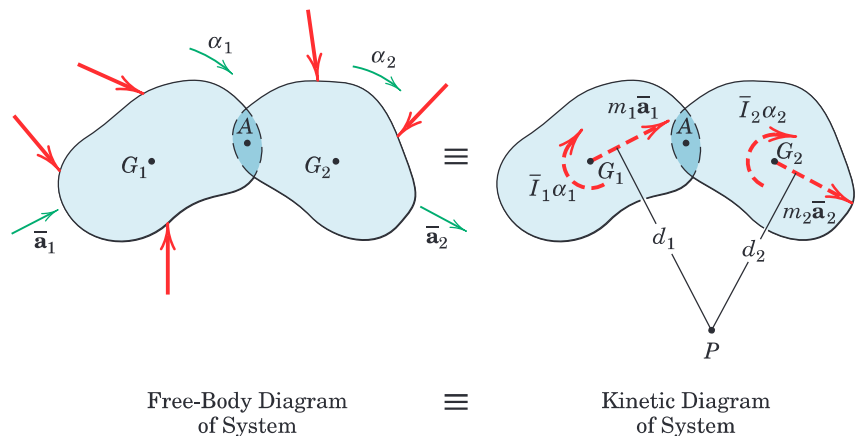


Figure 6/7

where the summations on the right-hand side of the equations represent as many terms as there are separate bodies.

If there are more than three remaining unknowns in a system, however, the three independent scalar equations of motion, when applied to the system, are not sufficient to solve the problem. In this case, more advanced methods such as virtual work (Art. 6/7) or Lagrange's equations (not discussed in this book*) could be employed, or else the system could be dismembered and each part analyzed separately with the resulting equations solved simultaneously.



KEY CONCEPTS

Analysis Procedure

In the solution of force-mass-acceleration problems for the plane motion of rigid bodies, the following steps should be taken once you understand the conditions and requirements of the problem:

1. Kinematics. First, identify the class of motion and then solve for any needed linear and angular accelerations which can be determined solely from given kinematic information. In the case of constrained plane motion, it is usually necessary to establish the relation between the linear acceleration of the mass center and the angular acceleration of the body by first solving the appropriate relative-velocity and relative-acceleration equations. Again, we emphasize that success in working force-mass-acceleration problems in this chapter is contingent on the ability to describe the necessary kinematics, so that frequent review of Chapter 5 is recommended.

2. Diagrams. Always draw the complete free-body diagram of the body to be analyzed. Assign a convenient inertial coordinate system and label all known and unknown quantities. The kinetic diagram should also be constructed so as to clarify the equivalence between the applied forces and the resulting dynamic response.

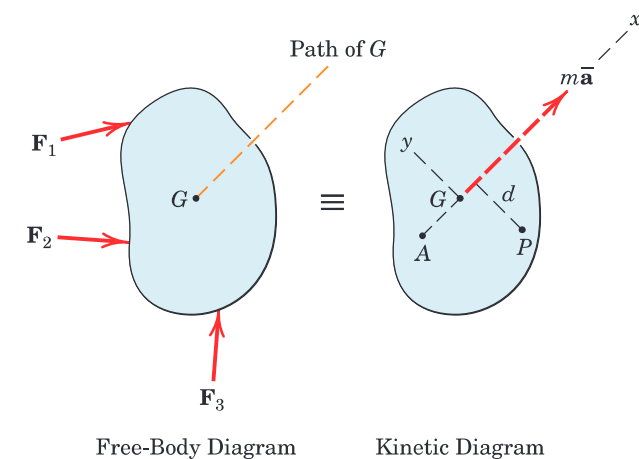
3. Equations of Motion. Apply the three equations of motion from Eqs. 6/1, being consistent with the algebraic signs in relation to the choice of reference axes. Equation 6/2 or 6/3 may be employed as an alternative to the second of Eqs. 6/1. Combine these relations with the results from any needed kinematic analysis. Count the number of unknowns and be certain that there are an equal number of independent equations available. For a solvable rigid-body problem in plane motion, there can be no more than the five scalar unknowns which can be determined from the three scalar equations of motion, obtained from Eqs. 6/1, and the two scalar component relations which come from the relative-acceleration equation.

*When an interconnected system has more than one degree of freedom, that is, requires more than one coordinate to specify completely the configuration of the system, the more advanced equations of Lagrange are generally used. See the first author's *Dynamics, 2nd Edition, SI Version*, 1975, John Wiley & Sons, for a treatment of Lagrange's equations.

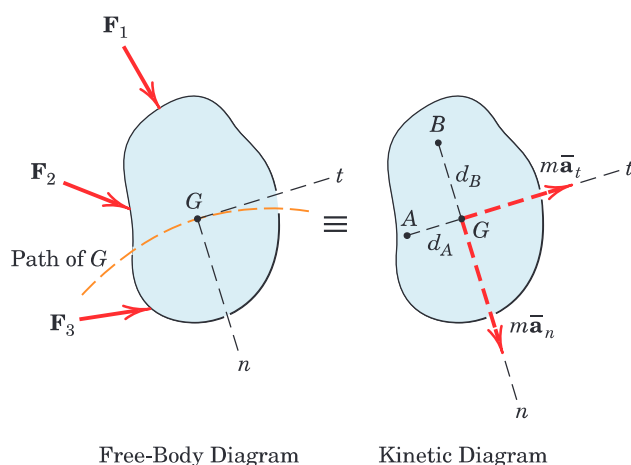
In the following three articles the foregoing developments will be applied to three cases of motion in a plane: *translation*, *fixed-axis rotation*, and *general plane motion*.

6/3 Translation

Rigid-body translation in plane motion was described in Art. 5/1 and illustrated in Figs. 5/1*a* and 5/1*b*, where we saw that every line in a translating body remains parallel to its original position at all times. In rectilinear translation all points move in straight lines, whereas in curvilinear translation all points move on congruent curved paths. In either case, there is no angular motion of the translating body, so that both ω and α are zero. Therefore, from the moment relation of Eqs. 6/1, we see that all reference to the moment of inertia is eliminated for a translating body.



(a) Rectilinear Translation
($\alpha = 0$, $\omega = 0$)



(b) Curvilinear Translation
($\alpha = 0$, $\omega = 0$)

Figure 6/8

For a translating body, then, our general equations for plane motion, Eqs. 6/1, may be written

$$\begin{aligned}\Sigma \mathbf{F} &= m\bar{\mathbf{a}} \\ \Sigma M_G &= \bar{I}\alpha = 0\end{aligned}\quad (6/6)$$

For rectilinear translation, illustrated in Fig. 6/8*a*, if the x -axis is chosen in the direction of the acceleration, then the two scalar force equations become $\Sigma F_x = m\bar{a}_x$ and $\Sigma F_y = m\bar{a}_y = 0$. For curvilinear translation, Fig. 6/8*b*, if we use n - t coordinates, the two scalar force equations become $\Sigma F_n = m\bar{a}_n$ and $\Sigma F_t = m\bar{a}_t$. In both cases, $\Sigma M_G = 0$.

We may also employ the alternative moment equation, Eq. 6/2, with the aid of the kinetic diagram. For rectilinear translation we see that $\Sigma M_P = m\bar{a}d$ and $\Sigma M_A = 0$. For curvilinear translation the kinetic diagram permits us to write $\Sigma M_A = m\bar{a}_n d_A$ in the clockwise sense and $\Sigma M_B = m\bar{a}_t d_B$ in the counterclockwise sense. Thus, we have complete freedom to choose a convenient moment center.



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The methods of this article apply to this motorcycle if its roll (lean) angle is constant for an interval of time.

6/4 Fixed-Axis Rotation

Rotation of a rigid body about a fixed axis O was described in Art. 5/2 and illustrated in Fig. 5/1c. For this motion, we saw that all points in the body describe circles about the rotation axis, and all lines of the body in the plane of motion have the same angular velocity ω and angular acceleration α .

The acceleration components of the mass center for circular motion are most easily expressed in n - t coordinates, so we have $a_n = \bar{r}\omega^2$ and $a_t = \bar{r}\alpha$, as shown in Fig. 6/9a for rotation of the rigid body about the fixed axis through O . Part *b* of the figure represents the free-body diagram, and the equivalent kinetic diagram in part *c* of the figure shows the force resultant $m\bar{\mathbf{a}}$ in terms of its n - and t -components and the resultant couple $\bar{I}\alpha$.

Our general equations for plane motion, Eqs. 6/1, are directly applicable and are repeated here.

$$\begin{aligned}\Sigma \mathbf{F} &= m\bar{\mathbf{a}} \\ \Sigma M_G &= \bar{I}\alpha\end{aligned}\quad [6/1]$$

Thus, the two scalar components of the force equation become $\Sigma F_n = m\bar{r}\omega^2$ and $\Sigma F_t = m\bar{r}\alpha$. In applying the moment equation about G , we must account for the moment of the force applied to the body at O , so this force must not be omitted from the free-body diagram.

For fixed-axis rotation, it is generally useful to apply a moment equation directly about the rotation axis O . We derived this equation previously as Eq. 6/4, which is repeated here.

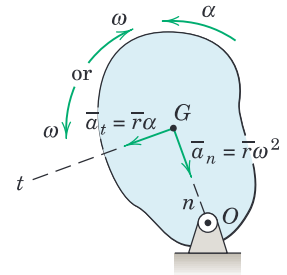
$$\Sigma M_O = I_O \alpha \quad [6/4]$$

From the kinetic diagram in Fig. 6/9c, we may obtain Eq. 6/4 very easily by evaluating the moment of the resultants about O , which becomes $\Sigma M_O = \bar{I}\alpha + m\bar{a}_t\bar{r}$. Application of the parallel-axis theorem for mass moments of inertia, $I_O = \bar{I} + m\bar{r}^2$, gives $\Sigma M_O = (I_O - m\bar{r}^2)\alpha + m\bar{r}^2\alpha = I_O\alpha$.

For the common case of rotation of a rigid body about a fixed axis through its mass center G , clearly, $\bar{\mathbf{a}} = \mathbf{0}$, and therefore $\Sigma \mathbf{F} = \mathbf{0}$. The resultant of the applied forces then is the couple $\bar{I}\alpha$.

We may combine the resultant-force component $m\bar{a}_t$ and resultant couple $\bar{I}\alpha$ by moving $m\bar{a}_t$ to a parallel position through point Q on line OG , Fig. 6/10, located by $m\bar{r}\alpha q = \bar{I}\alpha + m\bar{r}\alpha(\bar{r})$. Using the parallel-axis theorem and $I_O = k_O^2 m$ gives $q = k_O^2 / \bar{r}$.

Point Q is called the *center of percussion* and has the unique property that the resultant of all forces applied to the body must pass through it. It follows that the sum of the moments of all forces about the center of percussion is always zero, $\Sigma M_Q = 0$.



Fixed-Axis Rotation
(a)

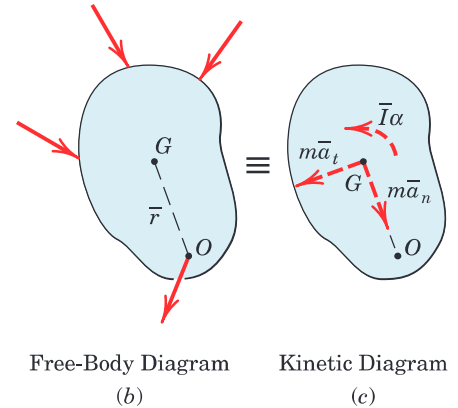


Figure 6/9

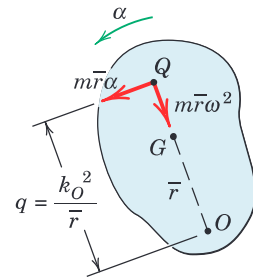


Figure 6/10

6/5 General Plane Motion

The dynamics of a rigid body in general plane motion combines translation and rotation. In Art. 6/2 we represented such a body in Fig. 6/4 with its free-body diagram and its kinetic diagram, which discloses the dynamic resultants of the applied forces. Figure 6/4 and Eqs. 6/1, which apply to general plane motion, are repeated here for convenient reference.

$$\begin{aligned}\Sigma \mathbf{F} &= m\bar{\mathbf{a}} \\ \Sigma M_G &= \bar{I}\alpha\end{aligned}\quad [6/1]$$

Direct application of these equations expresses the equivalence between the externally applied forces, as disclosed by the free-body diagram, and their force and moment resultants, as represented by the kinetic diagram.

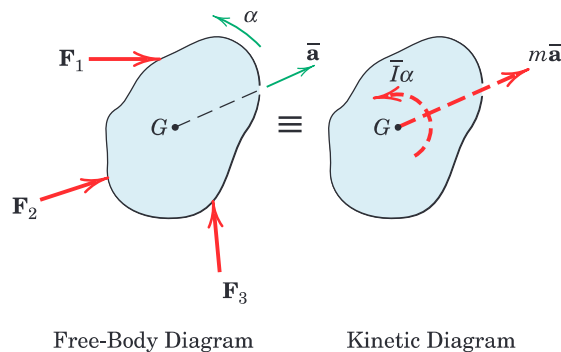


Figure 6/4, repeated



KEY CONCEPTS

Solving Plane-Motion Problems

Keep in mind the following considerations when solving plane-motion problems.

Choice of Coordinate System. The force equation of Eq. 6/1 should be expressed in whatever coordinate system most readily describes the acceleration of the mass center. You should consider rectangular, normal-tangential, and polar coordinates.

Choice of Moment Equation. In Art. 6/2 we also showed, with the aid of Fig. 6/5, the application of the alternative relation for moments about any point *P*, Eq. 6/2. This figure and this equation are also repeated here for easy reference.

$$\Sigma M_P = \bar{I}\alpha + m\bar{a}d \quad [6/2]$$

In some instances, it may be more convenient to use the alternative moment relation of Eq. 6/3 when moments are taken about a point *P* whose acceleration is known. Note also that the equation for moments about a

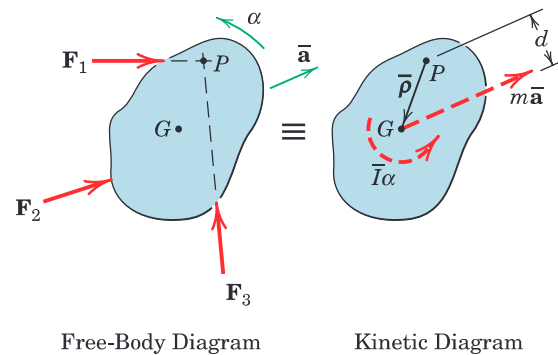


Figure 6/5, repeated

nonaccelerating point O on the body, Eq. 6/4, constitutes still another alternative moment relation and at times may be used to advantage.

Constrained versus Unconstrained Motion. In working a problem in general plane motion, we first observe whether the motion is unconstrained or constrained, as illustrated in the examples of Fig. 6/6. If the motion is constrained, we must account for the kinematic relationship between the linear and the angular accelerations and incorporate it into our force and moment equations of motion. If the motion is unconstrained, the accelerations can be determined independently of one another by direct application of the three motion equations, Eqs. 6/1.

Number of Unknowns. In order for a rigid-body problem to be solvable, the number of unknowns cannot exceed the number of independent equations available to describe them, and a check on the sufficiency of the relationships should always be made. At the most, for plane motion we have three scalar equations of motion and two scalar components of the vector relative-acceleration equation for constrained motion. Thus, we can handle as many as five unknowns for each rigid body.

Identification of the Body or System. We emphasize the importance of clearly choosing the body to be isolated and representing this isolation by a correct free-body diagram. Only after this vital step has been completed can we properly evaluate the equivalence between the external forces and their resultants.

Kinematics. Of equal importance in the analysis of plane motion is a clear understanding of the kinematics involved. Very often, the difficulties experienced at this point have to do with kinematics, and a thorough review of the relative-acceleration relations for plane motion will be most helpful.

Consistency of Assumptions. In formulating the solution to a problem, we recognize that the directions of certain forces or accelerations may not be known at the outset, so that it may be necessary to make initial assumptions whose validity will be proved or disproved when the solution is carried out. It is essential, however, that all assumptions made be consistent with the principle of action and reaction

and with any kinematic requirements, which are also called *conditions of constraint*.

Thus, for example, if a wheel is rolling on a horizontal surface, its center is constrained to move on a horizontal line. Furthermore, if the unknown linear acceleration a of the center of the wheel is assumed positive to the right, then the unknown angular acceleration α will be positive in a clockwise sense in order that $a = +r\alpha$, if we assume the wheel does not slip. Also, we note that, for a wheel which rolls without slipping, the static friction force between the wheel and its supporting surface is generally *less* than its maximum value, so that $F \neq \mu_s N$. But if the wheel slips as it rolls, $a \neq r\alpha$, and a kinetic friction force is generated which is given by $F = \mu_k N$. It may be necessary to test the validity of either assumption, slipping or no slipping, in a given problem. The difference between the coefficients of static and kinetic friction, μ_s and μ_k , is sometimes ignored, in which case, μ is used for either or both coefficients.



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Look ahead to Prob. 6/103 to see a special-case problem involving a crash-test dummy such as the one shown here.